Agent Harnesses

Four products shipped solo since April by designing the environment agents build in

Design Engineer · Postbyte LLC · April–November 2026

Finsum (live) · Langol (TestFlight) · Sinfon (beta) · Wondo (beta)

An agent harness is where my design decisions live so agents can't drift from them

Web scaffold: Flue agent harness over TanStack, Hono and Cloudflare, wrapped in Capacitor for mobile

Native scaffold: SwiftUI and Apple Intelligence for iOS and macOS

Each one is assembled from the skills and MCP servers its stack needs. Claude plans, Mimo builds and reviews, Claude Design prototypes, and Figma holds what survives.

A prototype takes between a weekend and a week.

Specs are the design artifact: each capability is a spec, and the spec changes in the same commit as the behaviour

- `CLAUDE.md` names each repo's source of truth
- `openspec/specs/`: one spec per capability. Langol has 24 and Sinfon has 33
- `openspec/changes/` holds work in flight, with delta specs
- When code and spec disagree: find out which is wrong, and fix both together

Finsum removes the app: your weekly budget is a WhatsApp thread with undo

- Set a weekly baseline, text "12 lunch", and every reply shows what's left
- "undo" reverses any spend. "@" lists your goal and debt funds
- The week closes itself, and the recap arrives with your next message
- No bank login: Finsum knows only what you send it

Live at finsum.app · free, limited launch group

Vikram Rojo · Design Engineering Case Study

Langol keeps every inference on the device, so the whole app works in airplane mode

- Grading, speech, translation and between-session coaching all run on-device with FoundationModels
- System frameworks only, with no third-party dependencies in the app target
- Apple is used only for storage and delivery: Background Assets for lessons, private CloudKit for progress
- Every intelligence feature has a fallback: self-assessment without Apple Intelligence, typing without the microphone

One codebase, two App Store products: Langol (Romance) and Langol ESL

Langol lets Swift decide what's decidable and the model only what isn't: 0 false positives in 260 judgements

- Gender agreement and conjugation are decided in Swift from bundled data. The model's category list doesn't contain them
- The checker abstains whenever it isn't sure
- Tested on 500 native Tatoeba sentences per language, unmodified and perturbed
- Spanish: 0 false positives, 0 missed, 0 spurious. Other languages are reported as unmeasured, not perfect

Sinfon

A quarter-tone sketching app where LilyPond engraves and the app owns the music

- 24-TET: maqam and Western scales on one grid, for Mac and iPad
- The file format is a strict subset of LilyPond, so every sketch is a real score
- Scales guide, never block: out-of-scale notes are advice, never refusals
- Imports 30 of 32 Mutopia LilyPond files, 29 of them clean

Beta · 1.0 macOS build signed and packaged

Sinfon hosts its own MCP server, so Claude edits the open sketch through the app's undo stack

- Claude's edits use the same edit commands as the grid and inspector
- Each Claude edit is one undo step, with harmony-follow and phrase rules applied
- "Allow Claude to edit open sketches" is off by default. Claude can always read
- The Activity tab marks every change as yours or Claude's, with Undo

Wondo was the first proof: a two-sided marketplace built by descoping to fit the harness

- Interactive-fiction marketplace: authors publish in Ink and readers browse, read, tip and subscribe on mobile
- Emdash CMS did double duty as platform and AI harness. If a feature could be descoped to fit it, it was
- Stripe Connect means authors own their stores
- Hardest engineering: making sign-in and checkout feel like one surface across providers

Beta at wondo.app

Four products shipped in seven months by one person

Shipped

- Wondo: public beta, wondo.app
- Finsum: live, finsum.app, limited launch
- Langol: TestFlight, two App Store products
- Sinfon: beta, signed 1.0 macOS build

What carried across

- Two scaffolds instead of four stacks
- Spec-first, with QA before every commit
- Trust boundaries written as architecture
- Store, privacy and licensing work done in-house

Design leadership moves upstream: decide what the environment makes impossible

- Data that can't leave the device
- A model that can't judge what a table can decide
- An agent that can't edit outside the user's undo stack

What I'd do differently: instrument outcomes as early as correctness.